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Fault structure, wear mechanisms and rupture processes in pseudotachylyte generation

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ABSTRACT

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Fault-generated pseudotachylyte is found within both cataclastic and mylonitic host rocks suggesting that rapid catastrophic displacements have occurred at a variety of depths within paleoseismogenic zones. Pseudotachylyte-bearing fault zones represent a composite of structural features associated with the process of earthquake rupture propagation and coseismic slip. The development of multiple pseudotachylyte veins in fault linkages, duplexes, sidewall ripouts, en echelon arrays and brittle zones suggests repeated rupturing during a series of characteristic earthquakes. Each earthquake, as a coseismic slip event, can be subdivided into initial rupture, acceleration, stable sliding and final deceleration stages. These evolve through distinctive sequences of wear and deformation mechanisms that vary with sliding velocity, duration of slip, total displacement and the hydrodynamics of the developing fault zone. Slip is thought to proceed toward surface refinement and possible frictional melting following the propagation of leading shear fracture process zones. The passage of the initial process zone of oblique fracturing would be followed by linkage to a throughgoing structure with asperity reduction through brecciation, comminution and refined cataclastic flow for frictional melting in an abrasive wear-dominant model. At greater depths in the presence of mylonitic anisotropy, slip would proceed through initial layer-parallel surfaces and duplex linkages with rapid surface refinement through plastic smearing and laminar flow for frictional melting in an adhesive wear-dominant model.

Introduction

Pseudotachylyte is a friction melt developed during brittle faulting (Allen, 1979; Maddock, 1983; Grocott, 1981; Spray, 1987). The occurrence of pseudotachylyte within fault zones is interpreted as the fossil remnant of paleoseismic events (Sibson, 1975, 1989). Pseudotachylyte is commonly found within inactive fault zones now exhumed by erosion where brittle ruptures propagated at depths within the crust approximating the former seismogenic zone. The occurrence of pseudotachylyte along particular faults suggests that the associated fractures were related to dynamic rupture propagation and slip during incremental coseismic displacement. The process of earthquake rupture propagation within the seismogenic zone is inherently complex at all scales, but a lack of adequate geologic data inhibits any detailed investigations into the actual process of rupture and slip (Scholz, 1990). The study of pseudotachylyte-bearing fault structures, however, can provide, by direct observation, a unique view of the process of earthquake rupturing that cannot be obtained through indirect seismic studies.

This paper will present some of the distinctive fault structures associated with the occurrence of pseudotachylyte in a strike-slip fault setting, based on studies in southern Maine and previously published literature, and make some speculations as to the processes at work during earthquake rup-

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turing. The discussion of fault structure, wear mechanisms and rupture processes in a single paper will attempt to bridge the gap between the field/microscopic observations of fault zones and the theoretical/experimental realm of fault mechanics. This will help to build a conceptual dynamic model for the generation of pseudotachylyte and provide possible directions for further research. An understanding of the sequence of rupture processes involved in faulting from an experimental approach (Cox and Scholz, 1988a,b; Jiefan et al., 1990; Shimada and Cho, 1990) or from a geologic perspective, as in this study, will be invaluable in modelling source zones associated with dynamic earthquake rupturing.

Fault rock associations

Pseudotachylyte can be found within formerly active fault zones now exhumed by erosion. Reported occurrences include both ductile mylonitic as well as brittle cataclastic host-rock associations. These exposures effectively span the entire depth range of the strike-slip seismogenic zone for typical shallow continental crustal faults (Fig. 1).

The ductile mylonitic regime develops friction melts that are reworked by continued plastic shearing. These exposures display features of concurrent brittle and plastic deformation (not simply the reactivation of earlier ductile fault structures), expressed as intermittent brittle rupturing within a background of continuous plastic shearing. Pseudotachylytes from the mylonitic regime have been reported from mylonitic gneisses of the Barthélémy Massif in the French Pyrenees (Passchier, 1982, 1984), the Outer Hebrides Thrust of northwest Scotland (Sibson, 1980a), the Musgrave block in central Australia (Wenk and Weiss, 1982) and from Late Paleozoic mylonites of the southern coast of Maine (Swanson, 1988). Pseudotachylyte veins in these exposures show evidence of plastic deformation along with the adjoining host rock and the development of internal foliations during shear. Flattened, recrystallized porphyroclasts and mineral aggre-

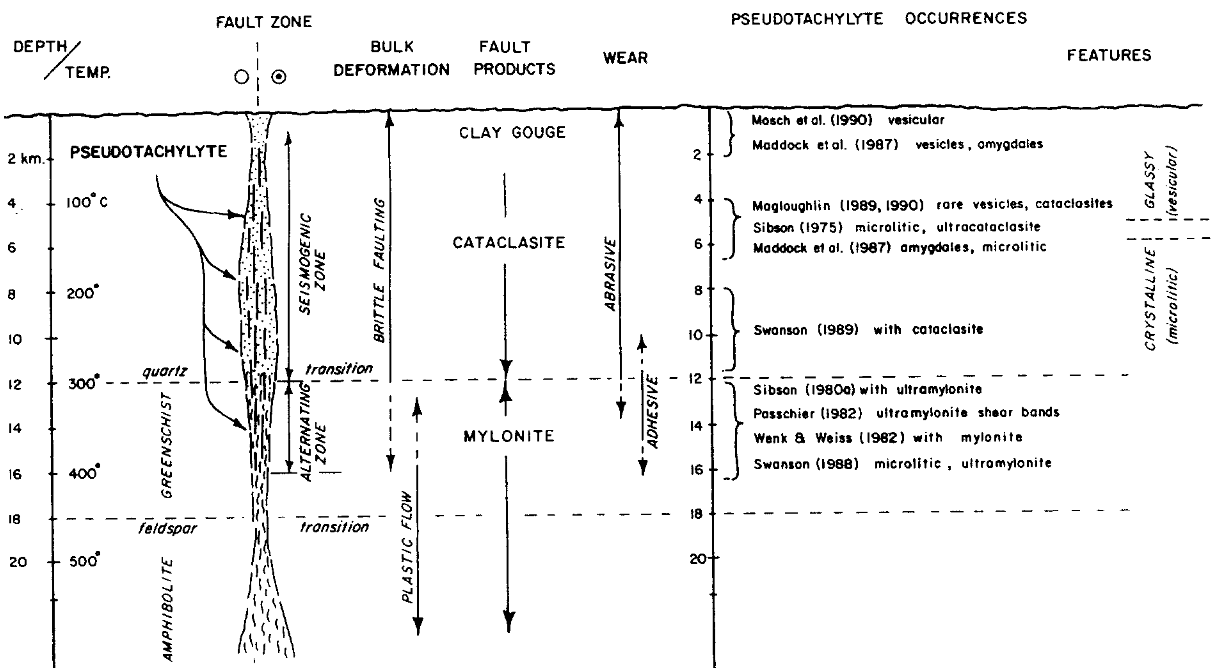


Fig. 1. Profile through a conceptual strike-slip seismogenic zone, after Sibson (1983) and Scholz (1988), showing the brittle–plastic transition, variation of deformation and wear mechanisms with depth in the crust, and the distribution of selected pseudotachylyte occurrences (some from non-strike-slip sources) within both mylonitic and cataclastic fault zones.

gates are aligned parallel to these internal fabrics. Injection veins are flattened and rotated. Sibson (1980a) suggests that these veins reflect transient discontinuities in the ductile flow field. The plastic deformation of pseudotachylyte injection veins in mylonite regimes offers evidence for truly concurrent brittle–plastic deformation.

The brittle cataclastic regime develops friction melts in conjunction with the active cataclasis of the adjoining wall rocks. A close spatial and temporal association of pseudotachylyte and the development of cataclastic layers has been reported from the Nason Terrane of Washington (Magloughlin, 1989), the Hidaka metamorphic belt in Japan (Toyoshima, 1990) and in association with Late Paleozoic faults in coastal Maine (Swanson, 1989b). The pseudotachylyte in these exposures typically contains clasts of older cataclasite and itself occurs as clasts in younger cataclasite. Multiple melt–cataclasite cycles reported by Magloughlin (1989, 1990, 1992) clearly establishes a relationship with cataclasis and related processes of abrasive wear in the brittle deformation regime.

Pseudotachylyte can commonly develop due to progressive strain hardening exhibited by many fault zones during uplift and erosion, thereby superimposing cataclastic features on earlier mylonitic ones (Hanmer, 1989). Single exposures may exhibit a progression of structures formed throughout the history of uplift and cooling from the brittle–plastic transition with continual aseismic plastic shearing through intermittent seismic slip, to repetitive cataclastic rupturing in the brittle deformation regime in the upper levels of the former seismogenic zone.

Nature of the seismogenic zone

The distribution of pseudotachylyte at the time of its formation within a fault zone corresponds to the seismogenic zone of the crust. Current models for a typical strike-slip seismogenic zone include the brittle–plastic transition in the crust as playing a key role in the structure of the seismogenic zone and in the development of earthquake ruptures (Fig. 1). The early models of Sibson (1983, 1989) proposed a basic crustal framework of an upper frictional seismogenic section dominated by faulting (or brittle behavior) and a lower, generally aseismic section dominated by plastic deformation. The boundary between the two sections is delineated by the transition from brittle to plastic deformation mechanisms in quartz at about 300°C and 12 km depth for an undisturbed geothermal gradient. Higher heat flow and locally steeper geothermal gradients along faults would elevate this transition to shallower levels in the crust (Sibson, 1983; Scholz, 1990).

The extrapolation of the high-temperature flow law for the lower crust to intersect the lower-temperature pressure-dependent linear friction law in the upper crust is, however, an over-simplification (Scholz, 1990). Variable responses to deformation in polymineralic crustal rocks argue against such a simple transition. Because of this, Scholz (1988) introduced a broad transitional "semi-brittle" zone to account for the variation in deformation mechanisms of quartz and feldspar in quartzo-feldspathic crustal rocks, which is not fully-plastic until the feldspar transition temperature at about 450°C and 18 km depth. The bulk rock will assume the properties of the dominant mineral component and thus the brittle–plastic transition is gradational over several kilometers. This semi-brittle transition zone corresponds, approximately, to depths in the crust typical of the greenschist facies.

The lower section of the seismogenic zone has been described as an "alternating zone" (Scholz, 1988), with bulk semi-brittle flow that alternates with coseismic dynamic slip, due to downward rupture propagation. The association of pseudotachylyte with mylonitic host rocks (Sibson, 1980a; Passchier, 1982, 1984; Swanson, 1988) as an example of this alternating zone suggests intermittent rupture propagation into this quasi-plastic mylonitic flow regime at depth (Sibson, 1980a; Strehlau, 1986; Hobbs et al., 1986; Scholz, 1988). It also clearly shows the slip- or strain-rate control on deformation and wear mechanisms that develops in this domain during actual coseismic slip as opposed to interseismic creep.

However, considerable bulk "ductile" deformation also occurs in the upper crustal levels within rocks that typically deform by brittle faulting. Such alternating deformation can be seen associated with cataclastic faults where accompanied by continual aseismic solution-assisted creep (i.e. diffusive mass transfer; Knipe, 1989) in the adjacent wall rock and by cataclastic flow within developing cataclasites or through minor fault arrays. Power and Tullis (1989), for example, describe shallow, brittle faults that display evidence for alternating coseismic cataclasis and aseismic creep during slickenside development as part of an on-going seismic cycle.

Friction and wear during slip

Friction during slip within the upper crust is dependent on the dynamic adhesion of a population of asperity contacts between the sliding surfaces (Rabinowicz, 1965; Bowden and Tabor, 1973; Scholz, 1990). Surface damage during sliding results in wear due to the interlocking and ploughing of asperities (Engelder, 1978). The localized high stresses at these asperities lead to localized brittle fracturing (abrasive wear) and/or plastic shearing (adhesive wear) with the possible development of localized high temperatures leading to incipient frictional melting (Logan, 1990). A large fraction of the energy expended during sliding is expressed as frictional heating (Engelder, 1978). The deformational response of the asperities as the mechanisms of wear during progressive surface refinement controls the frictional resistance during slip. Surface refinement progresses with wear of contacting asperities, increasing the real area of contact between the sliding surfaces (Teufel and Logan, 1978).

An important change in the dominant wear mechanisms operating during slip in a conceptual strike-slip fault zone also occurs within the brittle–plastic transition zone (Scholz, 1988, 1990), from abrasive-dominant wear above ~12 km depth to adhesive-dominant wear below and ultimately to continuous adhesion through fully-plastic deformation at depths greater than ~18 km (Fig. 1). The abrasive wear domain is characterized by brittle behavior and unstable frictional, velocity-weakening slip with the fracturing of asperities, development of loose wear particles and the production of a cushion of cataclasite. The adhesive wear domain is characterized by semibrittle behavior and stable frictional, velocity-strengthening slip with plastic deformation of asperities and material transfer to opposing surfaces during slip. This surface refinement during slip produces a deformed layer of processed asperities that may, ultimately, lead to shear heating and frictional melting as the surfaces evolve. Frictional melting can be an effective mechanism of surface refinement that rapidly leads to a maximum real area of contact by the melting and redistribution of asperities during slip.

The generation of pseudotachylyte by frictional melting is, therefore, a wear mechanism related to the evolution of frictional shear resistance during coseismic slip. The amount of heat generated during faulting is also dependent on the sliding velocity and slip duration, as well as the confining pressure and the availability of fluids, since these affect the normal stress on the fault during sliding. The frictional properties of fault surfaces must change dramatically throughout each slip event as the surfaces evolve. Friction will vary from static to lower dynamic values upon development of a throughgoing surface, and may drop suddenly due to melting and thermal pressurization of the fault zone. These expected variations in friction can, in turn, be correlated with sequences of wear and deformation mechanisms that develop during dynamic rupture propagation. The detailed nature of such a rupture event may be deciphered through the examination of structures bearing evidence of frictional melting.

Typical features associated with pseudotachylyte

Many studies of pseudotachylyte occurrences have focused on friction melts where preserved in structural reservoirs. The sites of generation, however, are commonly along narrow seams that represent the actual asperity contacts during slip. Both the reservoir structures and generating surfaces are created during the coseismic slip event when the rupture processes that initiate slip eventually lead to frictional melting. Clues to the nature of these rupture processes are preserved in the fault–fracture structures associated with pseudotachylyte, and in the microstructures and textures of the former melt itself.

Conditions during frictional melting

Friction melts begin to develop sometime during each slip event, given sufficient slip velocity and duration and, once produced, typically survive as melt through the duration of slip. The final cooling and solidification of the friction melt, combined with any subsequent alteration or deformation, are responsible for the overall matrix textures and microstructures in the pseudotachylyte. Due to the nature of frictional heating that produces pseudotachylyte, each slip event is associated with a transient coseismic thermal pulse in the immediately adjacent wall rocks (Scholz, 1990). A time/temperature plot for an idealized coseismic thermal spike can be used to model the evolution of the friction melt and its solidification to pseudotachylyte (Fig. 2). This evolution begins with an early frictional heating stage associated with the processing of asperities during surface refinement that leads to melting. Wall rocks are "flash" melted and, in some cases, superheated during shear. Peak melt temperatures have been estimated to be in excess of ~ 1000°C by Sibson (1975) and Maddock (1983) based on host-rock melt relations and over 1520°C by Wenk and Weiss (1982) based on SiO_2 glass compositions. McKenzie and Brune (1972) hypothesized that temperature increases of over 1000°C could be reached during geologic faulting. Experimental results on a friction welding machine by Spray (1987) show this to be a rapid and catastrophic process with intense incandescence of the wall rocks during flash melting and estimated temperatures of ~ 1400°C. Maximum temperatures achieved during frictional sliding at sandstone asperities, using thermal dyes, was found to be ~ 1180°C (Tuefel and Logan, 1978)

Melt lifetimes may range from microseconds to several minutes (or hours to days) depending on slip velocity, slip duration and reservoir dimensions (Sibson, 1975) because these parameters affect the total heat generation and the rate of conductive heat loss to the adjacent wall rocks.

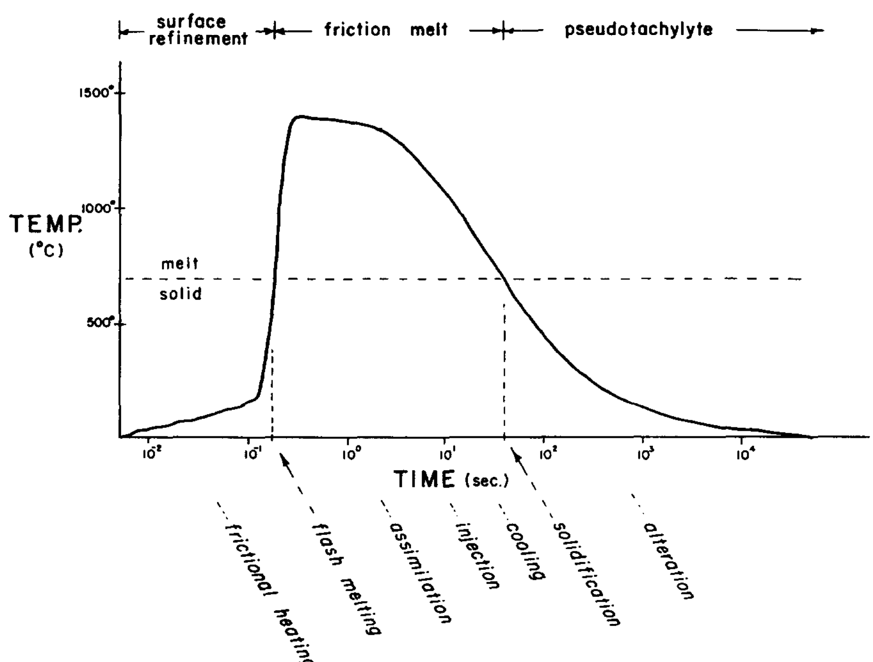
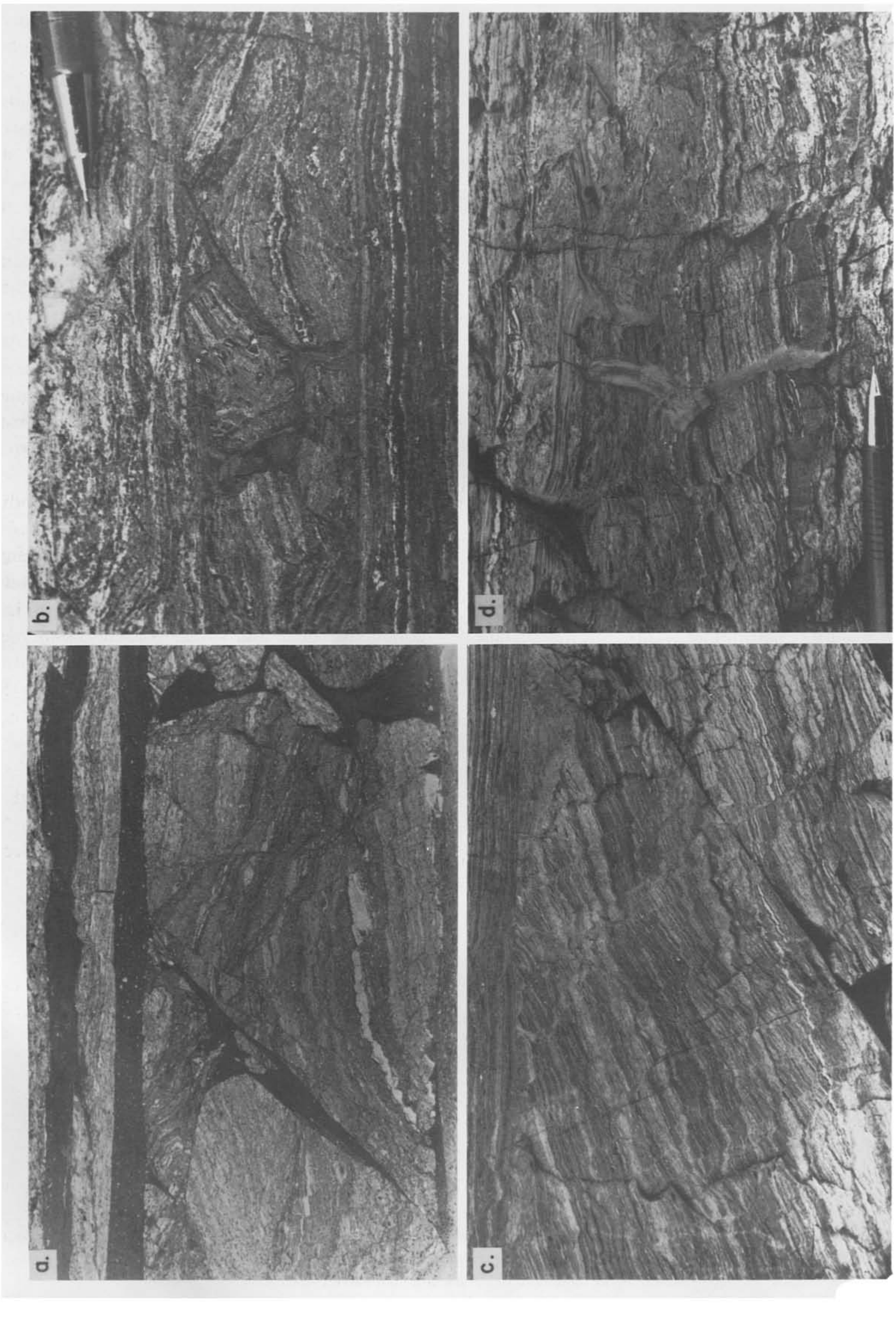


Fig. 2. Time/temperature curve for a hypothetical transient coseismic thermal spike associated with frictional melting and the production of pseudotachylyte. Various features of the surviving pseudotachylyte can be attributed to different stages in the temporal development of the friction melt and its solidification.


b.
d.
a.
c.

The melt solidifies during post-seismic quiescence, but contains remnant features related to processes associated with the immediately preceding slip event. While glassy veins and chill margins suggest rapid solidification, microlitic textures indicate sufficient time was available for slow, static crystallization. There is evidence in some examples for assimilation of fragmented wall rock during the melt phase with the development of haloes (Swanson, 1988; Maddock, 1983) and rounding of wall rock xenoliths (Sibson, 1975), as well as vapor exsolution in vesicles (Maddock et al., 1987). Geochemical studies suggest a direct melting source (Maddock, 1986, 1990), mixing of multiple parent lithologies (Magloughlin, 1989) and a preferential progressive melting sequence of hydrated mafic minerals followed by feldspar and quartz (Maddock, 1990). Selective melting may be indicated by a lack of mafic minerals as xenocrysts in the pseudotachylyte. Ragged fault vein contacts on the micro-scale may also result from preferential melting of mafic minerals in the adjacent wall rock (Spray, 1992-this issue).

The energy dissipation levels expected during seismic faulting as investigated by Moore and Sibson (1978) and Sibson (1980b), can lead to extensive thermal fragmentation of the wall rocks accompanying frictional melting. Fragmentation of the wall rocks and thermal spalling of the xenoliths would also facilitate assimilation. This may be a significant source of mineral and wall rock clasts in some pseudotachylyte. This process may also produce anomalously thick pseudotachylyte veins charged with clasts in varying states of assimilation.

Flow features commonly found within pseudotachylyte can be attributed to simple shear flow during displacement of the wall rocks along fault veins and injection flow between walls of extension fractures into dilational injection vein reservoirs. Deformed amygdales reported by Maddock et al. (1987) for certain Greenland pseudotachylyte were interpreted in terms of shear flow in a highly viscous cooling melt. Fault vein structures produced by viscous shear in the cooling friction melt could have potential as kinematic indicators of the sense of slip.

Deciphering the ambient and transient conditions during the slip event in terms of confining pressure, temperature, slip velocity, dynamic shear strength and particularly depth are more problematic. Glassy pseudotachylyte may be dominant above 5 km (Allen, 1979). The presence and mineralogy of vesicles and amygdales, representing the exsolution of a vapor phase from the melts under low lithostatic pressures, have also been used to infer a shallow (1.6 km) depth of faulting for the production of the Greenland pseudotachylytes (Maddock et al., 1987). Highly vesiculated pseudotachylyte has been found in the surface environment associated with landslides (Masch et al., 1985; Masch, 1990). Crystalline pseudotachylyte with microlitic textures appears to develop at deeper crustal levels (Sibson, 1975; Maddock, 1983). The association of plastically-deformed pseudotachylyte in mylonitic zones also indicates much deeper crustal levels (10–15 km) and higher ambient temperatures (300–450°C). Evidence suggests, then, that frictional melting can occur at all crustal depths within the seismogenic zone (Fig. 1).

Fault structures and associated dilatant reservoirs

Pseudotachylyte-bearing faults and associated vein arrays represent the development of genera-

Fig. 3. Pseudotachylyte from the Fort Foster Brittle Zone, southern Maine: (a) full thin-section photomicrograph (4 cm across) of multiple fault-parallel pseudotachylyte veins developed along mylonitic layering. Note the millimetre-wide generation zones, internal faulting, vein injections and older vein at top showing both ductile and brittle deformation effects; (b) outcrop photo (8 cm across) of coupled rupture surfaces showing a pseudotachylyte matrix to a localized fault breccia; (c) layer-parallel fault vein in upper part of photo (40 cm across) truncates rotated fault blocks in lower part of photo, suggesting considerable wear at the sliding contact; (d) outcrop photo (12 cm across) of layer-parallel fault vein and lateral injection vein showing internal flow layering and brittle offsets related to subsequent rupture events.

tion-surface/reservoir-structure systems during successive earthquake events (Figs. 3 and 4). The resulting fault vein and injection vein assemblages offer insight into the role of dynamic rupture propagation in fault zone evolution.

Fault vein and injection vein arrays

Pseudotachylyte is most commonly found in fault veins and injection veins (Figs. 3a,d and 4a) as initially recognized by Sibson (1975). The fault veins are typically a few millimeters to a few centimeters in thickness and may show variations in thickness related to irregularities in the fault surface. Injection veins of pseudotachylyte (Figs. 3a,d and 4a) are the most common reservoir for generated melts. Such veins typically lead the melt away from any generating surfaces into the cooler wall rocks. Injection veins may reach several centimetres in thickness and several meters in length. A high-angle or near-orthogonal relationship to the associated fault veins is common. The type II vein classification of Magloughlin (1989) incorporated isolated fault veins with associated lateral injection veins and classified various fault vein types into systems that show progressions toward multiple coupled arrays. Sibson (1975) attempted to quantify the vein thickness-fault displacement relationship for single-jerk faults in the Outer Hebrides Thrust. Field measurements show a linear relationship with vein thicknesses proportional to displacement.

Pseudotachtylyte generation zones

Distinctive structures associated with pseudotachylyte include paired slip surfaces that isolate tabular zones of host rock (Figs. 3a,b and 4b), first recognized in the brittle zones of the Ikertoq belt in west Greenland (Grocott, 1981). These distinctive parallel fault configurations, called pseudotachylyte generation zones, are defined by pairs of overlapping layer-parallel slip surfaces that serve as the dominant displacement structures. The fault-bounded slabs between these overlapping surfaces commonly exhibit a complex strain history. Internal fracture assemblages consisting of orthogonal dilatant veins and conjugate shear fractures indicate fault-parallel extension associated with the injection of pseudotachylyte.

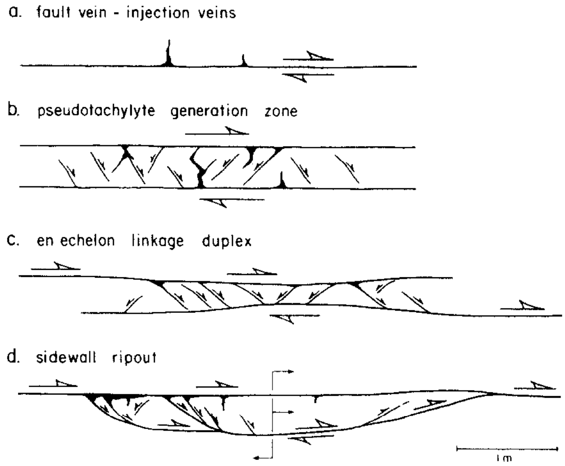


Fig. 4. Fault structures associated with pseudotachylyte generation: (a) fault vein/injection vein systems; (b) paired shear systems of pseudotachylyte generation zones (after Grocott, 1981); (c) en echelon linkage duplex in typical extensional geometry (Swanson, 1988); (d) sidewall ripouts with leading and trailing structural assemblages (Swanson, 1989b).

A variety of vein arrays have been described by previous workers (Sibson, 1975; Park, 1961), many of which have two parallel bounding fault veins in the array. The type III veins of Magloughlin (1989) also resemble the paired shears of Grocott (1981), while the ladderworks and pull-aparts of Sibson (1975) are similar except that there is less overlap between fault surfaces.

Strike-slip slab duplexes

Detailed mapping of tabular structures in mylonitic exposures of the Fort Foster Brittle Zone of southern Maine (Fig. 4c) has shown these paired structures to be elongate areas of extensive overlap between the ends of en echelon strike-slip fault segments (Swanson, 1988). Internal deformation within the tabular zones, by conjugate faulting between overlapping generating slip surfaces, serves as the mechanism of displacement transfer and finite strain accommodation between the coupled fault segments during slip. This style of deformation and fault linkage creates distinctive pseudotachylyte-bearing, slab-shaped, strike-slip duplexes, particularly in highly anisotropic rock. Extensional and contractional geometries of internal fracturing within the fault-bounded slabs depend on the sense of slip and stepping direction between the overlapping fault segments. Contractional duplexes tend to thicken with displacement through internal imbrication. Extensional duplexes with severe listric fault rotations (Fig. 3c) may thin catastrophically and lead to the formation of breccia with pseudotachylyte matrix (Fig. 3b).

Sidewall ripouts

Similar structures associated with pseudotachylyte in both the mylonitic and cataclastic fault zones of southern Maine as well as in western Greenland (Fig. 4d) consist of coupled extensional and contractional ramps (Swanson, 1989b) that define tabular to plano-convex fault lenses adjacent to the dominant slip surfaces. These structures, termed "sidewall ripouts", are interpreted to be mesoscale examples of adhesive wear that were generated as tabular ripouts up to 35 m or more in length during slip along the main fault. Adhesion of the fault blocks during slip ruptures the adjoining fault wall, ripping out a slab of host rock and translating it along during displacement. This ripout lens, at least temporarily, acts as an asperity (Scholz and Engelder, 1976), plowing its way through the adjoining wall rock along the leading contractional ramp until the cessation of slip or it is sheared off during continued displacement. Translational plowing causes distortion of the fault surfaces at the leading ramp and creates an extensional reservoir at the trailing ramp for pseudotachylyte produced along the dominant slip surfaces. The intersection of the extensional splay faults with the principal slip surfaces creates a local structural reservoir due to rotated-block movement along these listric trailing ramp surfaces. With significant displacements, tapered zones of pseudotachylyte-matrix breccias develop in the ripout tails as described from Greenland (Swanson, 1989b). Similar melt-dominant zones with a plano-convex geometry and showing a high degree of assimilation, described by Magloughlin (1989), may have a similar origin. Plano-convex sidewall ripouts (Swanson, 1989b) are similar to wear features found in metals (Rabinowicz, 1965; Bowden and Tabor, 1973). Since adhesive wear depends on the slip velocity and slip surface refinement between the slipping surfaces, sidewall ripouts most likely develop during the later stages of each associated slip event as the velocity and real area of contact along the fault increase.

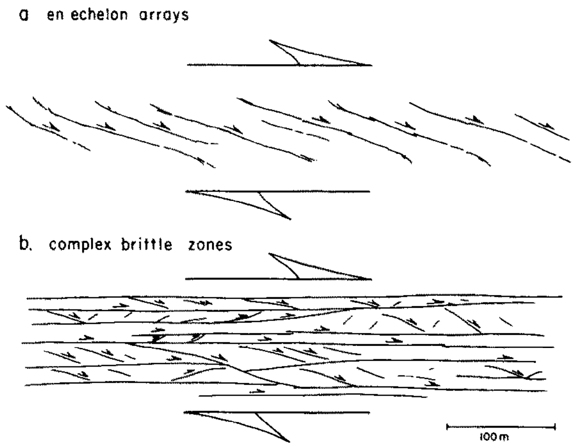


Fig. 5. Structural settings associated with pseudotachylyte fault veins: (a) shear bands and en echelon R-shear arrays as precursors to through-going fault zone develoment; (b) complex brittle zones of repeated layer-parallel ruptures.

Multiple fault vein arrays

Several occurrences of multiple pseudotachylyte fault vein arrays are found in distinctive structural settings when viewed at a larger scale (Fig. 5). These multiple vein arrays show evidence for repeated rupturing with identical fault styles and deformation mechanisms for successive earthquake events.

En echelon arrays

Evidence for intermittent coseismic slip is apparent within some shear systems that have evolved as R-shear, en echelon arrays. These may occur as mesoscale shear-bands in a zone of distributed regional fabric (Fig. 5a). Individual shear elements occur as oblique slip surfaces or fault zones that develop localized pseudotachylyte which, therefore, must be coseismic. These oblique slip surfaces are subject to re-orientation during continued shearing toward lower angles relative to the shear direction. Passchier (1982) describes oblique pseudotachylyte fault veins that, with continued plastic deformation, evolved into lower-angle ultramylonite shear bands. Swanson (1989c) has described oblique cataclastic fault zones with minor pseudotachylyte in an en echelon R-shear array on the kilometer-scale in association with centimeter-scale shear band fabrics in the host lithologies. Pseudotachylyte is locally developed in linking P-shear ramps between segments. With the continual accumulation of finite strain, the development of these mesoscale shear bands and en echelon arrays may be a precursor to the evolution of a larger-scale, throughgoing fault structure. Large-scale en echelon arrays that evolve with reorientation to lower angles relative to the shear direction may lead to the formation of complex brittle zones described below.

Brittle zones

Pseudotachylyte veins are commonly found in well-defined zones of intense shear fracturing up to several hundred meters in width and particularly within anisotropic host rock (Fig. 5b). Zones of pseudotachylyte-bearing brittle fractures in the Ikertoq belt of west Greenland (Grocott, 1981) occur as complex, sub-parallel, overlapping arrays up to 100 km in length. These complex arrays of multitudes of fault-parallel pseudotachylyte veins were called "brittle zones" (Grocott, 1981; Swanson, 1988). The Fort Foster Brittle Zone in southern Maine consists of hundreds of rupture surfaces in a 60-m-wide zone of intense fracturing. The brittle zone itself appears to have a paired shear or duplex structure, with slip localization occurring along the outer boundary zones.

The repeated rupturing apparent in these brittle zones suggests a history of paleoseismic activity. The structural similarity of these repeated ruptures is due to the strong structural control of the host rock anisotropy and a consistent style of fault segment linkage. This provides a clear structural mechanism for the Schwartz and Coppersmith (1984) model for the evolution of so-called "characteristic earthquakes".

Rupture model

Rupture at a point

Each coseismic slip event is typically recorded on strong motion accelerometers as a sudden ground acceleration to coseismic slip velocities, of

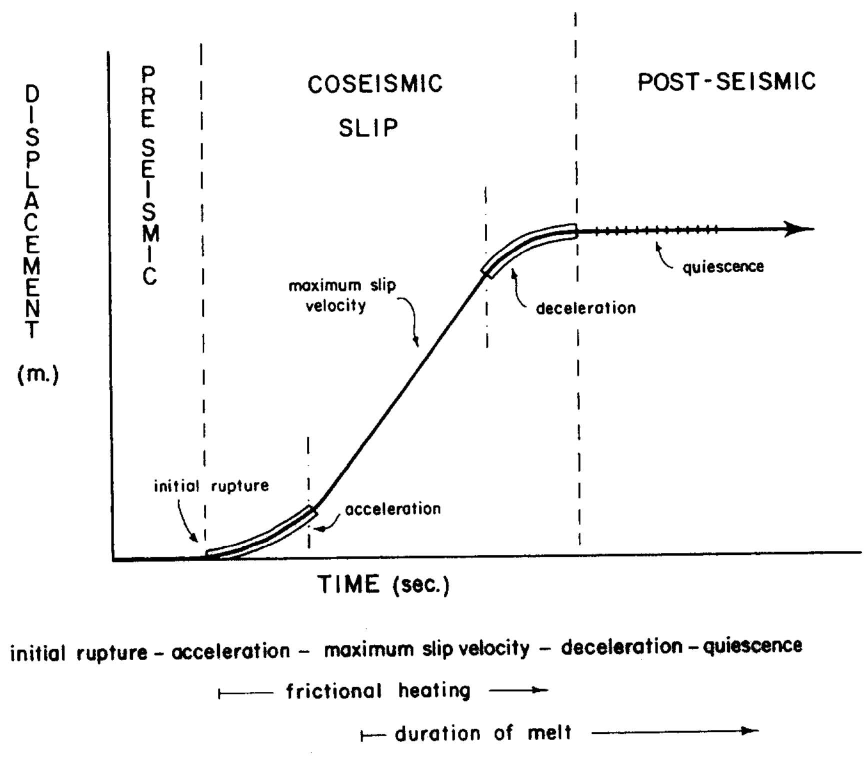


Fig. 6. Rupture model for coseismic slip at any particular point on a slip plane, depicted on hypothetical displacement versus time curve showing processes associated with initial rupture, linkage, acceleration, maximum slip velocity, deceleration and post-seismic quiescence.

the order of several meters per second, followed by deceleration until the fault blocks are again at rest (Sibson, 1989; Scholz, 1990). On a local scale, the actual rupture and displacement processes associated with coseismic slip and earthquake generation are complex, with deformation and wear mechanisms being dependent on pressure and temperature (Scholz, 1988) and strongly influenced by slip velocity (Rabinowicz, 1965; Shimamoto, 1989; Spray, 1989). Deformation and wear mechanisms at any crustal level will, therefore, vary continuously with the evolution of slip velocity and frictional resistance during each coseismic event.

Stages in the development of coseismic slip

An instructive view of the coseismic slip event is the corresponding displacement versus time plot where the slope reflects slip velocity and changes in slope reflect acceleration and deceleration (Fig. 6). Each slip event can be subdivided, at least hypothetically, into an initial acceleration stage leading to a maximum slip velocity, followed by a deceleration stage until the cessation of slip. Total slip duration for each event may be only a few seconds.

Acceleration involves the propagation and linkage of an initial fracture array resulting in the clearing of asperities and surface refinement facilitating unrestrained slip. The maximum slip velocity stage follows after surface refinement through various wear mechanisms. Given sufficient displacement, velocity and slip duration along these surfaces, frictional heating may lead to pseudotachylyte generation. Deceleration and post-slip quiescence would be associated with the injection, cooling and solidification of the friction melt. This succession of stages also corresponds to a progression of wear and deformation mechanisms that develop along the fault surface during the accumulation of displacement and the build-up and dissipation of frictional heat.

Shear heating that leads to partial or complete melting could produce a dramatic reduction in dynamic shear strength (Jaeger, 1942; McKenzie and Brune, 1972; Cardwell et al., 1978; Sibson, 1980a). Melt production may result in acceleration of slip due to wall rock lubrication and thermal pressurization of the fault by the melt. Melt pressures result from expansion of the host rock upon melting (e.g. up to 12% volume for rocks of tonalitic compositions, as reported by

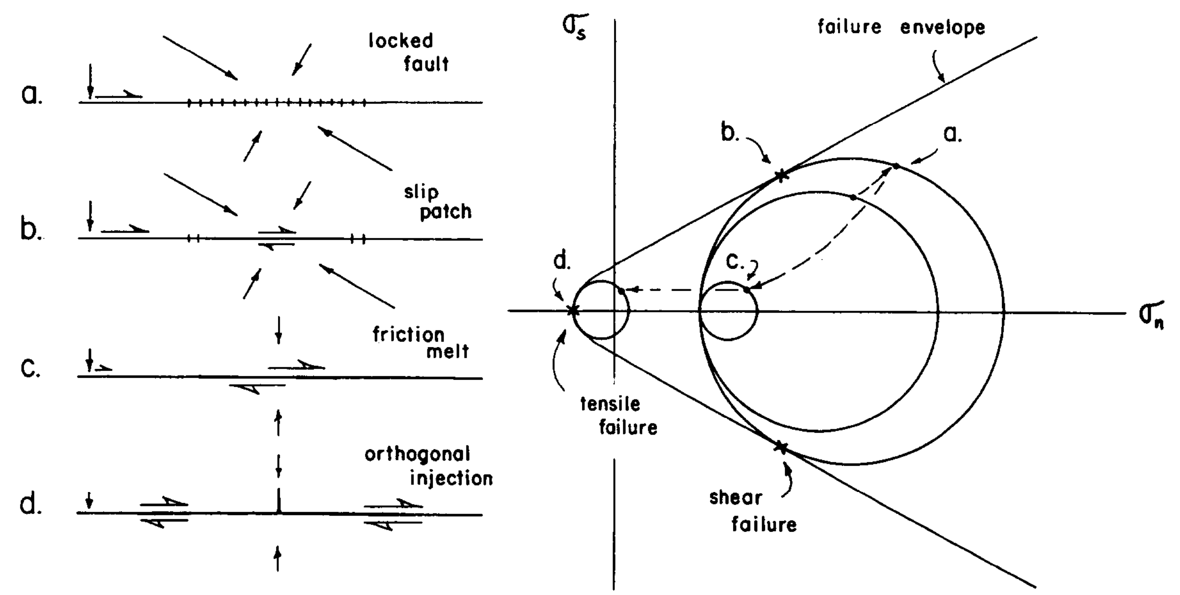


Fig. 7. Mohr diagram showing the generation of an orthogonal injection of melt during slip: (a) gradual increase in differential stress on locked fault zone; (b) failure through shear fracture propagation along fault; (c) coseismic stress drop to residual differential stress during unrestrained slip; (d) rapid decrease in effective normal stress upon generation of friction melt with associated hydraulic fracturing and orthogonal melt injection.

Maddock et al., 1987). High transient dynamic melt pressures on the order of 3 GPa may be responsible for the formation of a high P–T Al-rich pyroxene found in some Australian pseudotachylytes according to Wenk and Weiss (1982). Orthogonal injection veins of pseudotachylyte projecting off of the dominant fault veins suggest lateral wall rock injection of pseudotachylyte that can be attributed to hydraulic fracturing related to these high transient melt pressures (Fig. 7). The orthogonal geometries reflect a drop in shear stress on the dominant slip surface (Fig. 7) associated with thermal pressurization and melt lubrication during slip as the friction melt layer is generated (Maddock et al., 1987; Sibson, 1973).

Extensive melt lubrication and pressurization along fault segments could modify slip velocity and acceleration, promoting high seismic efficiency and high dynamic stress drops (Sibson, 1973, 1977), thus driving rupture propagation to its farthest extent. Scholz (1990), however, points out that any silicate melts would have high viscosities because coseismic flash melting leaves little time for water to enter the melt. The resulting melt layer may then exhibit some shear strength, reducing the effects of melt lubrication. Preferential melting of hydrated mafic phases (Maddock, 1990), on the other hand, may concentrate water in early friction melts thus reducing viscosities. However, melt lubrication may tend to occur in localized patches at evolving asperity contacts, so that its role in reducing the effective dynamic strength over the entire fault plane may be limited (Scholz, 1990).

The lubricating effect of the melt and the pressurization of the fault zone will ensure low effective normal stress and low frictional shear resistance. This would, at least locally, dramatically decrease the frictional heat production at the generating sites during slip and allow the melt to begin to cool. The melt layer may then experience localized increases in viscosity and increasing frictional resistance during slip which together act as a possible source of intermittent adhesion. Significant viscosity changes during longer slip events could lead to cyclic rupturing during single events. This may be similar to successive melting–freezing cycles associated with power oscillations observed in the frictional melting experiments by Spray (1987). Multiple vein-parallel layering could be caused by multiple acceleration pulses during slip, perhaps related to melt production and evolution. Lateral wall rock injection of the melt as seen in outcrop would also cause a sudden loss of fluid pressure and its associated lubricating effects. This rapid loss of lubrication could then lead to localized adhesion, seizure of the fault blocks during slip and the development of the sidewall ripouts described above.

The commonly undeformed nature of most pseudotachylyte suggests that once produced, friction melts survive as a liquid throughout the deceleration stage until the cessation of slip and, upon cooling, statically weld the fault surfaces together. This appears to be an efficient strain-hardening process in that most pseudotachylyte fault veins are preserved intact with few that can be shown to have re-ruptured during subsequent slip events. Subsequent slip events rupture repeatedly through intact anisotropic host rock and create a collage of layer-parallel pseudotachylyte veins as seen in the brittle zones described above.

Each displacement increment, therefore, is likely to proceed through an initial, unstable frictional stage of acceleration with rupture, linkage and decoupling, to a maximum slip rate stage with the development of cataclastic and/or plastic flow, frictional heat production with pseudotachylyte generation, melt lubrication and thermal pressurization of the fault zone. This is followed by a deceleration stage and the development of late mesoscale adhesive wear features, then final post-slip quiescence and the solidification of pseudotachylyte veins.

The role of fluids

The presence of hydrous fluids within fault zones during slip can play an important role in controlling the dynamic shear strength and can buffer the production of friction melts (Sibson, 1973; Lachenbruch, 1980; Mase and Smith, 1985, 1987). Thermal pressurization of the fault zone due to fluids depends on the hydraulic characteristics of the host rock and the fault zone, the coefficient of friction, the slip velocity and the confining pressure. Theoretical work by Mase and Smith (1987) suggests that friction melting may still be possible in the presence of pore fluids providing that the permeability exceeds 10^{-15} m^{-2} with a coefficient of friction greater than 0.1 and slip velocities greater than 1 cm s^{-1}. This will allow frictional melting to occur before thermal pressurization of the fault zone (Mase and Smith, 1985). Given a low enough permeability, thermal fluid expansion, be it melt or water, due to frictional heating may lead to pressurization and rapid drops in the dynamic shear strength. However, an increase in permeability upon fault zone evolution could halt pressurization and cause a return to higher dynamic shear strengths (Mase and Smith, 1985).

A dynamic rupture model

The 1979 Imperial Valley earthquake

A useful dynamic model for earthquake rupture comes from the modelling studies of Archuleta (1984) and Quin (1990) for the 1979 Imperial Valley earthquake in California. This $M = 6.4$ earthquake was recorded on a collection of near-field strong motion accelerometers that allowed analysis of the propagation along the entire rupture length. These studies showed that the quake involved a 35-km rupture length with unidirectional propagation and maximum strike-slip displacements of 0.4 m. Strike-slip velocities ranged to 1 m s^{-1}, accompanied by abrupt accelerations and decelerations along the length of the rupture.

Propagating rupture front

The initiation of slip along a fault occurs progressively along a propagating rupture front. This is the point at which the adjacent fault blocks begin their acceleration to maximum slip velocities. The passage of this rupture front coincides with the development of initial rupture, linkage and throughgoing fault formation at any particular point. For the Imperial Valley event, average rupture velocity along the length of the fault was ~ 3 km s^{-1}. Maximum rupture velocities reached 9.8 km s^{-1}. The details of rupture initiation, linkage and acceleration may, theoretically, set the stage for eventual frictional melting during peak slip velocities.

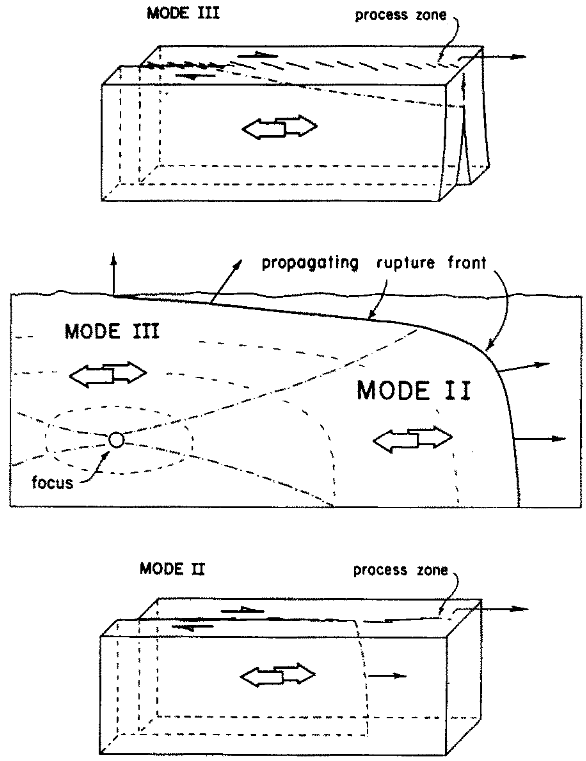


Fig. 8. Distribution of Mode II and Mode III rupture propagation and associated shear fracture process zones within a developing slip plane. For upper crustal levels dominated by cataclasis, faults evolve through a Mode III oblique en echelon process zone. At lower crustal levels in the mylonitic regime, faults evolve rapidly through an anisotropy-controlled Mode II shear fracture process zone.

Experimental studies of slip nucleation by Ohnaka and Kuwahara (1990) depict the rupture front to proceed from a nucleation phase of quasi-static, steady crack growth, to stable but accelerating crack growth at ~ 50 cm s^{-1}, to a critical length at which unstable dynamic rupture develops at 1–3 km s^{-1}. Stable crack growth during the breakdown process leads to transition and secondary rapid accelerations during unstable dynamic rupture propagation. Slip will only occur after local breakdown is completed. This creates a leading shear fracture "process zone" developed at the tip of the propagating rupture (Fig. 8). Deformation within the leading shear fracture process zone may correspond to the early stable fracturing under static friction conditions which leads to the transition to dynamic friction, linkage and unstable crack propagation. The transition from static to dynamic friction conditions occurs within the hypothesized critical slip distance which determines the dimensions of the leading process zone (Scholz, 1990).

Recent experimental work by Cox and Scholz (1988a,b) on shear fracture propagation and the associated process zone, points to a clear distinction between the style of shear fracture growth in Mode II and Mode III configurations (Fig. 8). Mode III propagation develops an oblique, en echelon array of tension cracks that precedes faulting (similar to the microscopic feather fractures of Friedman and Logan, 1970), while Mode II propagation is typically arrested by the growth of Mode I tension cracks. Mode II ruptures were found to only propagate under the influence of pre-existing anisotropy, as has been demonstrated by Laqueche et al. (1986) for slates. Mode II rupture propagation may be facilitated in the deeper parts of the fault zone under the influence of strain-induced anisotropy, such as anastomosing fault surfaces or a mylonitic foliation, thus promoting in-plane shear fracture propagation.

The development of an array of oblique Mode I tension fractures associated with Mode III shear fracture propagation leads to high initial wear rates and contributes to the formation of cataclasite (Cox and Scholz, 1988b). A similar mechanism for the development of cataclasite that proceeds through forward grain rotation and comminution has been proposed by Engelder (1974). This may be an effective mechanism at shallow crustal levels, but at deeper levels and higher confining pressures the dominant process zone mechanism would be oblique en echelon R-shear fracturing, as is commonly present in natural examples. At even deeper levels and in the presence of a pre-existing mylonitic anisotropy, overlapping sub-parallel *Y*-shears may be the dominant process zone mechanism (Swanson, 1989a).

Given the geometry of propagating rupture planes in a typical strike-slip setting, growth of the rupture surface will necessarily involve both modes of shear fracture propagation as illustrated in Figure 8. At near-focal depths in the seismogenic zone, the rupture plane propagates in the same direction as the slip vector as Mode II propagation. Mode II shear fracture propagation will be enhanced due to the prominent mylonitic foliation in the lower depths of the fault zone. At upper levels of the seismogenic zone above the focal depth, the rupture plane propagates upwards and perpendicular (or at least highly oblique) to the slip vector as Mode III propagation. This change in propagation mode with depth may also be reflected in the variation in the style of structural duplexing from lens to

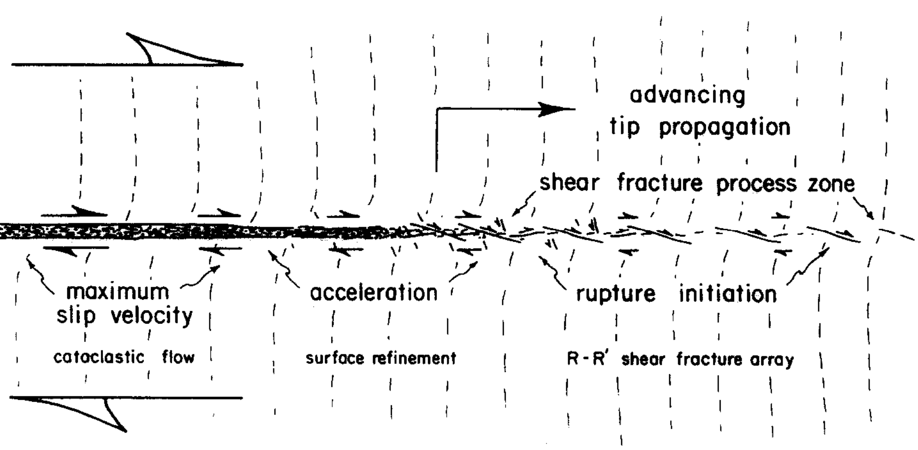


Fig. 9. Profile of the propagating rupture tip, the associated shear fracture process zone and stages in the development of coseismic slip.

long tabular overlap zones between fault segments in the presence of a mylonitic anisotropy (Swanson, 1989a).

Rupture tip profile

Rupture and slip can be examined from a more structural perspective with a telescoped view of a propagating rupture tip on the outer edge of the migrating slip patch associated with each coseismic slip event (Fig. 9). The propagating rupture tip develops as an advancing shear fracture process zone, perhaps associated with initial rupturing along an en echelon R-shear fracture array or oblique tension fracture array. A similar view of the shear fracture process zone and associated rupture tip has been presented by Cox and Scholz (1988a) based on interpretations from experimental results. *P*-shear linkages create a throughgoing rupture with acceleration proceeding as asperities are cleared by abrasive and adhesive wear, as well as by frictional melting during further surface refinement. Maximum slip velocities would be associated with the most evolved throughgoing structures where fault surfaces become cushioned by zones of cataclastic, plastic or viscous flow.

Using the sequence of deformational structures, processes and wear mechanisms expected during a discrete coseismic slip event, along with the hydrodynamic characteristics of the fault zone (Mase and Smith 1987), a conceptual model for dynamic rupture can be formulated. This conceptual model consists of a migrating fracture tip that leads expanding zones of distinctive structures, fault products and deformational processes developed during rupture propagation and slip (Fig. 9). The development of coseismic slip would follow the advancing rupture front behind a leading shear fracture process zone. As the rupture tip and process zone pass any point along the fault, the blocks will experience the rupture, acceleration, maximum slip rate and deceleration stages of a typical slip event as depicted in the static rupture model.

Slip window

While total rupture duration during the Imperial Valley earthquake was ~ 12 s, slip duration at any one point averaged only 1.2 s (Archuleta, 1984; Sibson, 1989): "Thus fault slip at any instant during rupture propagation was restricted to a band extending only a few kilometres behind the advancing rupture front" (Sibson, 1989, p. 5). Active dynamic slip along the fault would be "controlled by a moving window of rupture area" (Quin, 1990, p. 112).

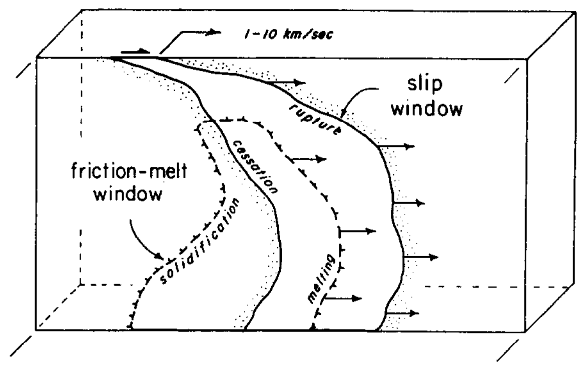


Fig. 10. Dynamic rupture model showing pseudotachylyte production and development within a moving friction melt window that trails the leading propagating rupture front. Melting during slip, after some surface refinement, marks the leading edge of the window. Solidification, sometime after the cessation of slip, marks the trailing edge of the window.

The leading edge of this moving slip window (Fig. 10) represents the actual rupture front during propagation. The deceleration stage leading to the cessation of slip defines the trailing edge of the slip window, which propagates at near-seismic velocities as does the leading rupture front (Scholz, 1990). This propagating "healing" front was initially modeled by Madariaga (1976) as a wave emanating back from the termination of rupture at the perimeter of the rupture plane. Based on the unidirectional rupture propagation in the Imperial Valley earthquake, slip is represented by a migrating slip window with the healing front trailing the propagation front to the perimeter of the rupture plane. The configuration of this slip window at any particular time during rupture reflects both the time of initial rupture and the duration of slip.

Friction-melt window

The point at which the first friction melt develops during slip marks the leading edge of a "friction-melt window" illustrated in Figure 10. This leading edge must follow some distance behind the propagating rupture front to allow time for rupture evolution to proceed to the appropriate level of surface refinement, slip velocity, slip duration and frictional heat generation. With the friction melt surviving after the cessation of slip, the trailing edge of the friction-melt window also follows some distance behind the back edge of the migrating slip patch, possibly interacting with a surge of post-seismic fluid flow (Sibson, 1981). The friction-melt window is shown in simplified form and assumes a homogeneous host rock, as well as uniform fault vein width and fluid activity with depth in the fault zone. The longer cooling times for crystalline pseudotachylyte at greater depths would alter this simplified model. The upper depth limit for melt production may be highly variable and controlled by the interaction of pore fluids and lower confining pressures. The lower boundary would extend into the plastic flow regime following the rupture depth limit for the seismogenic zone.

Thus, a dynamic friction melt window (Fig. 10) will trail the propagating rupture front throughout the duration of slip following the initial acceleration, and linger for some time after the cessation of slip. The configuration of this window at any given time during the rupture event will fluctuate in response to changes in slip velocity and duration, as well as the changes in rock type and hydrous fluids along the propagation path. The distribution of pseudotachylyte within the rupture plane may reflect the areas of greatest slip duration and highest slip velocity during the event. The passage of this window defines the area along the rupture capable of frictional melting, but it is ultimately the local structural setting and hydrodynamic conditions that control the stress configurations and magnitudes, the rupture and wear processes during slip and the possibility of producing friction melts.

Discussion

Wear sequences and surface refinement

The formation of pseudotachylyte along faults in brittle versus plastic regimes requires slightly different initial deformation mechanisms. Frictional melting during coseismic slip could develop through early abrasive wear and cataclasis at upper crustal levels, or through early adhesive wear and plastic flow in the mylonitic fault setting at lower crustal levels. Within a strike-slip seismogenic zone, slip would proceed from initial rupture toward surface refinement and frictional melting through two different sequences at different crustal levels. These sequences include an upper abrasive wear-dominant model and a lower adhesive wear-dominant model.

Abrasive wear-dominant model

The abrasive wear-dominant sequence (Fig. 11a) operates in the upper crustal levels of the strike-slip seismogenic zone. The abrasive wear sequence develops within active fault zones dominated by cataclasis and may involve Mode III shear fracture propagation. Initial process zones during rupture propagation would consist of oblique tension fracture arrays at shallow levels (Cox and Scholz, 1988a,b) and en echelon R-shear arrays at deeper levels. Surface refinement proceeds through forward clast rotation (Engelder, 1974) and comminution of the initiation breccia (Sibson, 1986), or through P-shear linkages in the en echelon array. Asperity reduction would be through brittle fracture, brecciation and comminution with high initial wear rates (Scholz, 1990). Continued slip leads to refined particulate flow within a cushion of cataclasite as it builds up along the fault surface. Grain-size reduction proceeds to some critical level where further strain becomes localized along oblique R-shears within the cataclasite layer, or along wall rock boundary shears (Engelder, 1978; Scholz, 1990). Pseudotachylyte in these active cataclastic fault zones tends to be thin fault veins sporadically developed along the margins of evolved cataclastic layers where shear strain has localized with high enough slip rates for frictional melting. Xenolith populations would include abundant cataclasite fragments. The intimate association of friction melts with cataclasis as outlined by Magloughlin (1990, 1992) would suggest that some pseudotachylyte may develop from a comminuted precursor, particularly at shallow crustal levels as in this abrasive wear-dominant model.

Once developed, these cataclastic layers may act as zones of continued weakness allowing slow interseismic creep. Strain would be accommodated by grain sliding and rotation, diffusion mass transfer and bulk cataclastic flow, as well as the passage of subsidiary rupture fronts. Deformation of the pseudotachylyte layers by rupture and the associated abrasive wear produces fragments of pseudotachylyte as part of the new cataclasite layers. Post-seismic fluid flow may also result in the complete cementation of the newly-developed fault rock (Stel, 1981, 1986) and alteration of any existing pseudotachylyte. Successive ruptures must then break through indurated cataclasite.

Adhesive wear-dominant model

The adhesive wear-dominant sequence (Fig. 11b) operates in the lower crustal levels of the strike-slip seismogenic zone. The adhesive sequence develops within active mylonitic fault zones that may be dominated by anisotropy-controlled Mode II shear fracture propagation. Rupture and slip proceed from initial layer-parallel *Y*-shear surfaces in leading process zones to overlap duplex linkages. The reactivation of the pre-existing planar anisotropy during rupture in these rocks provides a near-planar slip surface with few initial asperities and low initial wear rates during slip. Rapid surface refinement with a transition

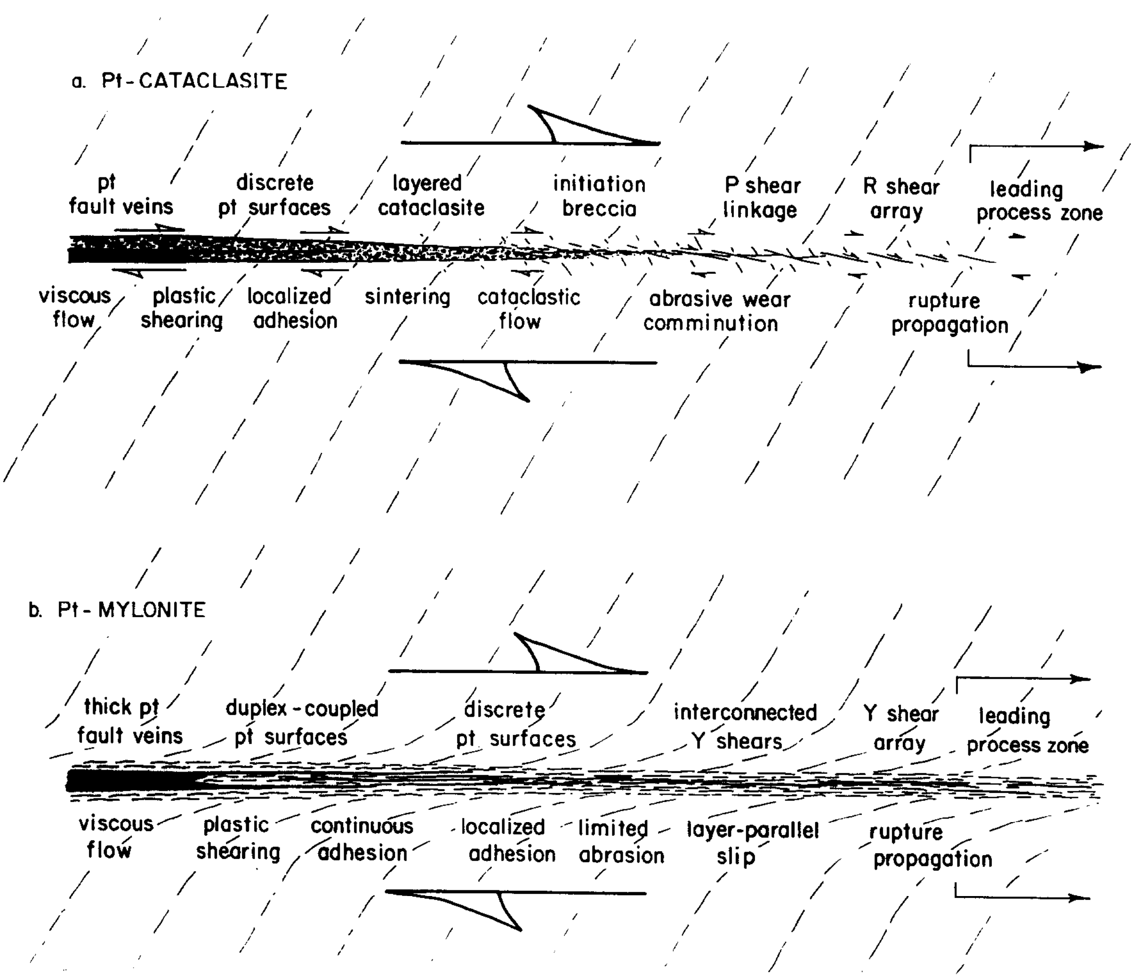


Fig. 11. (a) Upper crustal, abrasive wear-dominant model for frictional melting based on the pseudotachylyte (Pt)-cataclasite association, showing the progression of structures and wear mechanisms at cataclasite zone boundaries and internal oblique shears. (b) Lower crustal, adhesive wear-dominant model for frictional melting based on the pseudotachylyte (Pt)-mylonite association, showing progression of structures and wear mechanisms along foliation-parallel slip surfaces.

to total adhesion, as the real area of contact approaches the total area, leads directly to plastic smearing and laminar plastic flow without the extensive development of cushions of cataclasite. The surface refinement process is greatly accelerated thereby enhancing adhesion, plastic flow and frictional melting during slip which results in a much greater potential for the generation of pseudotachylyte. Larger, relatively xenolith-free fault veins and injection vein systems may be prominent. Evidence for initiation breccia (Sibson, 1986), cataclasis and abrasive wear will be rare. Further plastic deformation in the mylonites typically reworks the solidified melt veins where they become part of the host rock for the next series of ruptures. An intimate association between pseudotachylyte and ultramylonite may also be prevalent where early formed pseudotachylyte veins have been plastically-deformed and recrystallized to bands of ultramylonite (Sibson, 1980a; Passchier, 1982).

Fault zone evolution

The occurrence of pseudotachylyte within certain fault zones and in specific structural settings ties those structures to dynamic rupture propagation during paleoseismic events. Earthquake rupturing can now be viewed as an important structural process that is a key factor in the cumulative evolution of fault zones (Sibson, 1989). The presence of pseudotachylyte along faults enables the distinction to be made between those seismic structures resulting directly from dynamic rupture propagation and aseismic structures that develop through plastic shearing, cataclastic flow or small-increment, cumulative displacements; all of which contribute significantly to fault zone evolution.

Pseudotachylyte is clearly found within fault zones in various stages of their evolution. Sharp, small-scale brittle faults typically undergo frictional melting after only a few centimeters of slip. Initial en echelon fault arrays and linking structures can both develop friction melts. Friction melts can also be found in more evolved fault zones due to strain localization as cataclasite layers accumulate displacement, and during transient discontinuities in flow for well-developed mylonite zones. Both assemblages of fault rocks provide a clear example of the structural control of strain-induced anisotropy in providing avenues of layer-parallel slip for incremental ruptures. Thus, pseudotachylyte generation depends on an initial level of fault surface development, but can occur at any time during fault zone evolution from the first increment to the latest slip increment contributing to finite fault zone evolution.

Conclusions

The generation of pseudotachylyte by frictional melting along faults is an extremely rapid and dynamic process where physical conditions such as frictional shear resistance, fluid pressure, temperature and stresses are found to fluctuate over wide ranges in the second to microsecond time scales. Pseudotachylyte generated during coseismic slip is produced, developed and solidified within a moving friction melt window that trails the propagating slip patch during passage of the earthquake rupture. Pseudotachylyte is produced along generating slip surfaces after sufficient surface refinement and is preserved in fault vein and injection vein reservoirs. Pseudotachylyte shows evidence of a complex melt history from flash melting to final solidification, with evidence of both viscous shear and injection flow, assimilation of xenoliths, formation of crystallites and vesicles and subsequent plastic and brittle deformation. Transient melt pressures may exceed lithostatic pressures with thermal pressurization and melt lubrication of the fault zone. Rupture and slip in cataclastic faults of the upper seismogenic zone develop through Mode III shear fracture propagation and dominantly abrasive wear with slip concentration occurring at wall rock boundaries to produce friction melts. Rupture and slip in mylonitic fault zones of the lower seismogenic zone develop through Mode II shear fracture propagation and dominantly adhesive wear. Layer-parallel ruptures in mylonitic zones have a greater potential for pseudotachylyte generation due to the higher real area of contact during slip initiation that allows total adhesion and rapid plastic shear heating with little preliminary abrasive wear.

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